INCH-POUND

A-A-59101A
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SUPERSEDING
A-A-59101
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# COMMERCIAL ITEM DESCRIPTION
# WATTMETER
# METER, AUDIO POWER 20 Hz to 20 kHz

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

## 1. SCOPE

1.1 Description. This Commercial Item Description (CID), describes Wattmeter (Audio Power Meter). This CID is meant as a minimum requirement for the audio power meter in which only those manufacturers that meet or surpass the following requirements are qualified per this CID. This wattmeter is used to measure the output power of audio devices in the frequency range of 20 Hz to 20kHz.

## 2. CLASSIFICATION

2.1 Classification. Equipment covered by this CID shall be commercially available equipment and may modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800.

## 3. SALIENT CHARACTERISTICS

3.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

3.2. Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 115 volts ac (±10%) at line frequencies of 50 Hz to 60 Hz. Power consumption shall not exceed 25 watts.

3.3. Temperature Requirements.

3.3.1 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0° C to +50° C.

3.3.2 Nonoperating Temperature. The equipment shall meet its performance and accuracy requirements in a nonoperating environment of -10°C to +60° C.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1640.

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3.4 Mean Time Between Failures (MTBF). The equipment Mean Time Between Failures (MTBF) shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

3.5 Pre-conditioning Requirement. Each deliverable wattmeter shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

3.6 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours

3.7 Performance Characteristics.

3.7.1 Frequency Range. The minimum frequency range of the peak power meter shall be 20 Hz to 20 kHz.

3.7.2 Power Range. The power meter shall have as a minimum, range from 300 microwatts to 20 Watts.

3.7.3 Power Accuracy. The minimum power accuracy of the audio power meter shall be ±7.5% Full Scale Deflection (FSD).

3.7.4 Impedance. The impedance of the audio power meter shall be at least 10 ohms to 600 ohms.

3.7.5 Impedance Accuracy. The impedance accuracy of the audio power meter shall be at least ±7% for 70 Hz to 5 kHz.

3.7.6 Interface. The audio power meter shall be front panel controlled.

3.7.7 Connectors. The audio power meter shall have three binding posts on the front panel. Two of the binding posts shall be for inputs, and one binding post shall be for chassis ground.

3.7.8 Front Panel Display. The audio power meter shall have a meter/display on the front panel that displays power reading in linear mode – milliwatts/Watts (dependent on range selected) and in logarithmic mode – dBm.

3.7.9 Power Range Switch. The audio power meter shall have a switch on the front panel for the selection of an appropriate power range for unit under test power output measurement.

3.7.10 Impedance Selection Switch. The audio power meter shall have a switch on the front panel to select impedance (load) that will be applied to the unit under test.

3.7.11 Weight. The audio power meter shall not weigh more than 20 lbs.



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4. REGULATORY REQUIREMENTS. In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazard from the recovered materials.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product Conformance The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units provided they fall within the tolerances using conversion tables contained in the latest revision of Federal Standard 376, all other requirements of this Commercial Item Description.

5.3 Exceeded Metric Products. If a product is manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5.4 Bid Sample. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein.

6. PACKAGING.

5.1 Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing labeling shall be as specified in the contract or order.

7. NOTES

7.1 Source of Documents

7.1.1 Government documents Unless otherwise indicated, copies of specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave. Building 4D, Philadelphia, PA 19111-5094.

7.2 Previously Qualified Products. The following product(s) are known to have previously met the requirement of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of the CID and submit to bid testing and certification as required.

| Manufacturer | Model | NSN |
|---|---|---|
| Quadtech | 1840-9701 | 6625-00-937-6156 |
| Marconi | 893C | 6625-01-282-9397 |
| Avionics Specialist, Inc. | ASI-83C | 6625-01-441-9555 |



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